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Markerless Pose Estimation and Explainable Machine Learning for Assessing Kicking Technique and Shooting Performance in Youth Football Players

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Featured Application

A field workflow combining synchronized high-frame-rate video, markerless pose estimation, sports radar, and interpretable machine learning can support objective assessment of kicking mechanics and ball-velocity profiles in youth football without a laboratory marker-based motion-capture system.

Abstract

Markerless pose estimation offers a practical route to field-based biomechanical assessment, but its ability to predict distinct dimensions of football kicking performance remains uncertain. This study evaluated 60 youth football players (48 male, 12 female; age 16.35 ± 1.19 years, range 14.31–18.28) from Romanian youth football programs, who contributed 1134 valid instep kicks across dominant- and non-dominant-leg and velocity- and accuracy-priority conditions. Ball velocity was measured by sports radar and shooting precision by continuous radial target error. Twelve prespecified markerless biomechanical variables were derived from synchronized multiview high-speed video and evaluated using participant-wise nested cross-validation; outer-fold predictions were generated only for players not used in model fitting or tuning. Velocity-priority trials increased ball velocity but also increased radial error, while non-dominant-leg kicks were slower and less accurate. For ball velocity, Elastic Net achieved an out-of-fold MAE of 1.19 m/s (95% CI 1.12–1.25), approximately 4.5% of the overall trial-level mean ball velocity (26.23 m/s), and R^2 of 0.80 (95% CI 0.76–0.83); more complex nonlinear models provided no material improvement. Kicking-foot velocity, knee-extension angular velocity, and approach velocity were the most stable predictors. In contrast, radial-error prediction remained near baseline (Elastic Net R^2 = 0.01, 95% CI −0.02 to 0.04). These findings support cautious use of the workflow for internally validated ball-velocity profiling in similar youth players, but not as a stand-alone tool for assessing shooting precision.

Keywords: football; kicking biomechanics; markerless pose estimation; Elastic Net; ball velocity; shooting accuracy

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1. Introduction

Kicking is one of the decisive technical actions in football, but performance should not be reduced to ball speed alone. Successful shooting requires the player to produce sufficient ball velocity while controlling spatial error. Studies that manipulate instructions

toward speed or accuracy show a clear task-dependent speed–accuracy trade-off in football kicking [1,2]. This makes ball velocity and continuous target error complementary outcomes rather than interchangeable measures of technical quality.

The maximal instep kick is produced through a proximal-to-distal sequence involving pelvis and hip motion, knee extension, and rapid foot acceleration. In youth players, kicking kinematics and performance change across development, and ball speed can discriminate playing status in some age groups [3]. In professional players, waveform-based hip and knee movement patterns have also been associated with kicking velocity, although relationships can differ by sex [4]. These findings support a multivariable approach rather than interpreting a single joint angle as a complete representation of technique.

Conventional three-dimensional motion capture provides detailed biomechanical information but is difficult to deploy routinely on a football pitch. Markerless computer vision offers a lower-burden alternative. A football-specific field study showed that automated markerless analysis can recover three-dimensional kicking kinematics with strong agreement for several positional variables [5], while broader reviews of sports pose estimation emphasize both its potential and its sensitivity to camera geometry, occlusion, movement speed, and validation procedures [6,7].

Artificial intelligence has already been used in football to recognize passes and shots and to classify football-specific activities from wearable signals [8,9]. The present question is different: whether markerless kinematic features measured during a kick can estimate continuous performance outcomes—ball velocity and radial target error—in players not used for model fitting. This moves the evaluation from event detection toward participant-aware assessment of field-derived biomechanics.

Youth football provides a particularly useful context because technique is still developing and football-specific skill remains trainable. Structured training models can improve passing execution in junior players [10], while lower-limb power-oriented programs can improve speed and explosive capabilities in adolescents [11]. Technology-supported work in youth and team-sport athletes further demonstrates the feasibility of integrating objective reaction-time, spatial-orientation, and motor assessments into training environments [12,13]. These studies motivate practical, field-compatible methods that preserve football specificity.

Football-specific markerless studies have primarily focused on recovering or validating kicking kinematics [5,14], whereas football AI studies have often focused on action recognition rather than continuous biomechanical-performance estimation [8,9]. Accordingly, the specific knowledge gap is whether a compact markerless biomechanical feature set can internally generalize to unseen players for two distinct continuous outcomes: ball velocity and radial target error. Accuracy is conceptually broader than body-joint kinematics alone because foot–ball contact, ball direction, gaze/perception–action processes, and motor-control consistency may contribute to target error; these factors are not all captured by the present pose-derived features [1,2,15]. Repeated kicks from the same player create dependent observations, so row-wise machine-learning validation can substantially overestimate generalization. Athlete-level grouping and transparent reporting are therefore essential [16,17].

The present study addresses this gap by combining synchronized multiview high-speed video, markerless three-dimensional pose reconstruction, radar-measured ball velocity, continuous target error, repeated dominant- and non-dominant-leg trials, and explainable machine learning. The intended contribution is deliberately bounded: to characterize field-based kicking biomechanics and test internal participant-wise performance estimation in unseen players from the same study population. It is not presented as an externally validated or deployment-ready coaching system.

1.1. Aim, Hypotheses, and Exploratory Expectation

The aim of this study was to determine whether markerless pose-estimation features obtained during field-based football kicking were associated with, and could internally estimate, ball velocity and radial target error in youth football players. The directional hypotheses reflect the expected separation between proximal-to-distal speed-generation variables and spatial-control variables described in football-kicking biomechanics, while H2 remains comparatively exploratory because direct evidence for continuous radial target error is limited [2,15].

- H1. Greater kicking-foot velocity and knee-extension angular velocity will be positively associated with ball velocity.
- H2. Support-foot position, trunk orientation, and pelvis rotation will be more strongly associated with shooting accuracy than with ball velocity.
- H3. Multivariable machine-learning models will improve prediction relative to single-feature and simple linear baselines.
- H4. Non-dominant-leg kicks will show lower performance and greater kinematic variability than dominant-leg kicks.
- Exploratory expectation. Explainability analyses will identify a compact subset of kinematic features accounting for most of the predictive signal.

1.2. Novelty and Contribution

The same kick is represented by both an objective mechanical outcome (radar ball velocity) and an objective precision outcome (continuous radial target error), enabling the study to test whether the same biomechanical information predicts both dimensions equally well.

The markerless workflow is evaluated under field conditions, and its quality-control error is reported for event timing and selected biomechanical variables rather than assuming that pose-estimation output is the ground truth.

Machine-learning validation is grouped by participant, so the model must generalize to unseen players rather than merely to additional kicks from players already present in the training data.

Dominant/non-dominant leg and speed/accuracy instruction are incorporated into the experimental design, allowing the predictive analysis to be interpreted alongside an explicit football-specific speed–accuracy trade-off.

2. Materials and Methods

2.1. Study Design and Sample Size Rationale

The study used a randomized repeated-measures field experiment with 60 youth football players. Each player was scheduled to complete 20 kicks (five per leg $\times$ instruction condition), yielding a planned maximum of 1200 trial-level observations before quality exclusions. Trial-level observations were not treated as independent participants: participant identity defined the grouping unit in repeated-measures analyses and machine-learning validation. With 60 independent players, a conventional two-sided paired-measures calculation has approximately 80% power at $\alpha = 0.05$ to detect a standardized within-player effect of about $dz = 0.37$; this calculation pertains only to within-player contrasts and was not used to claim adequacy for the machine-learning analysis. No formal sample-size formula was used to guarantee stability of the four predictive models. The machine-learning component is therefore framed as an internally validated proof-of-concept, with the predictor set restricted a priori and participant-wise nested cross-validation used to estimate internal performance.

2.2. Participants

Participants were recruited from youth football programs in Romania. Eligibility included U16–U19 status, at least two years of systematic football training, and medical readiness to perform high-intensity instep kicks. Players with lower-limb pain or injury that limited kicking, incomplete return-to-sport clearance, or inability to complete the standardized protocol were not eligible for unrestricted testing. Sex, age, standing height, body mass, training age, playing position, competitive level, and dominant kicking leg were recorded. The final cohort spanned 14.31–18.28 years and included male and female players. Because only 12 participants were female, sex was not entered as a predictive covariate or used for sex-stratified models; any sex-associated confounding therefore remains possible and is addressed as a limitation.

The study was approved by the Research Ethics Commission of the Faculty of Physical Education and Mountain Sports, Transilvania University of Brașov (No. 814/16.02.2025), before testing. Written informed consent was obtained from adult participants; for all 54 participants who were minors at testing, written parent/legal-guardian consent and participant assent were documented on or before testing. The study-specific participant information described video recording and AI-based analysis.

2.3. Standardized Experimental Protocol

Testing was conducted using a stationary size-5 football at a fixed shooting distance of 11 m. Ball model and inflation pressure were recorded for every session. A 1.0 × 1.0 m target was positioned in the lower contralateral corner relative to the kicking leg, with the target center 1.0 m inside the relevant goalpost and 0.75 m above the ground. Right-foot kicks therefore targeted the left corner and left-foot kicks the right corner. Target and impact coordinates were recorded in centimeters in the goal plane; the continuous error measure was calculated relative to the trial-specific target center.

2.4. Randomization and Trial Procedure

Each player completed four five-kick blocks: dominant-leg velocity-priority, dominant-leg accuracy-priority, non-dominant-leg velocity-priority, and non-dominant-leg accuracy-priority. Block order followed a pre-generated balanced randomization schedule. In velocity-priority blocks, players were instructed to maximize ball speed while keeping the shot on goal; in accuracy-priority blocks, they were instructed to strike the predefined target with a high-intensity instep kick. Players used their preferred approach pattern within the marked run-up area. Trials were flagged invalid for slips, mishits, camera obstruction, missing performance measurements, a non-stationary ball, or protocol violations. Invalid attempts were retained in the audit log and repeated at the end of the relevant block when feasible. The standardized warm-up and scheduled recovery alone accounted for approximately 28–33 min per participant, excluding kick execution, setup, and repeated invalid attempts. The protocol therefore provided frequent recovery opportunities; no separate measure of momentary attention or mental focus was collected.

2.5. Video Acquisition, Calibration, and Synchronization

Four synchronized camera views recorded the kicking volume. The archive contained 5008 video files from 1252 attempted kicks. A source-file audit verified the encoded frame rate for every file: 3047 files were 240 frames/s and 1961 were 120 frames/s, all at 1920 × 1080. The mixed frame rates represented acquisition heterogeneity in the archived field recordings rather than a prespecified model factor. Because many trials contained mixed 120- and 240-frame/s views, each source file's own timestamp/frame-rate information was used, and the views were resampled/synchronized to a common time base before

multiview reconstruction and calculation of time-dependent biomechanical features. Resampling established a common time base but cannot recreate the finer temporal sampling absent from 120-frame/s recordings; therefore, event timing and derived velocities may be more sensitive to timing error when lower-frame-rate views contribute. Accordingly, the verified file-level metadata were treated as the timing authority rather than the session-level CC01/CC02 label.

2.6. Markerless Pose Estimation and Biomechanical Features

A multiview markerless workflow (Figure 1) was implemented using Pose2Sim v0.6.1 with the OpenPose BODY_25 keypoint model for two-dimensional detection and three-dimensional reconstruction [7]. Raw videos were retained separately from the analytic dataset, which used coded participant identifiers and derived kinematic variables. Events included support-foot contact, maximal backswing, ball contact, and early follow-through. The 12 primary predictors were prespecified by the investigators to represent approach mechanics, proximal-to-distal swing mechanics, trunk/pelvis orientation, and support-foot geometry, using variables supported by football-kicking biomechanics and recoverable with the markerless setup [3–5]. Other candidate variables were excluded to limit dimensionality relative to 60 independent participants. These quantities were continuous predictor measurements sampled at the specified events or windows; no categorical pose "hit" was inferred from knee, pelvis, trunk, or support-foot values.

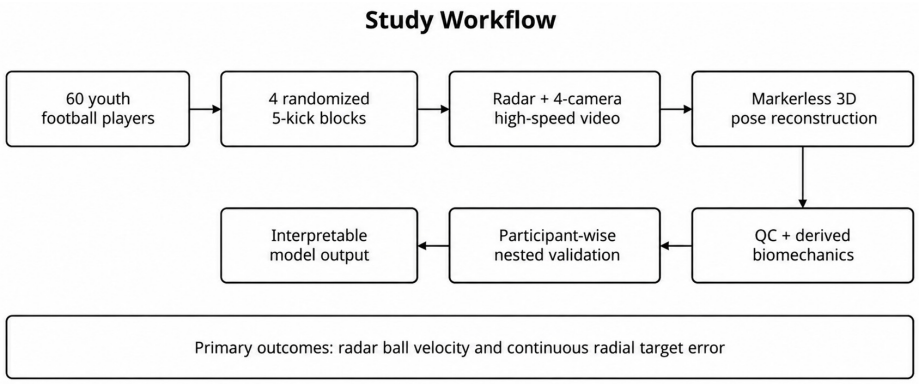


Figure 1. Study workflow from participant testing to markerless reconstruction, quality control, participant-wise model validation, and interpretable model output.

2.7. Shooting-Performance Outcomes

Ball velocity was measured and analyzed in m/s using a sports radar positioned outside the run-up corridor and aligned with the expected ball trajectory [18]. All inferential and machine-learning results use m/s; km/h was not used in the statistical or predictive analyses. Accuracy was quantified primarily as the Euclidean radial distance between the ball-impact point and the trial-specific target center in the goal plane: radial error = $\sqrt{[(\text{ximpact} - \text{xtarget})^2 + (\text{yimpact} - \text{ytarget})^2]}$, expressed in centimeters. For the secondary binary outcome, a target hit was defined geometrically as an impact occurring within ± 50 cm horizontally and ± 50 cm vertically from the center of the 1.0×1.0 m square target. The binary hit outcome was separate from, and not inferred from, the pose-derived predictors. Radial error remained the preferred accuracy outcome because it preserves information lost in a binary hit/miss label.

2.8. Quality Control

Pose-estimation confidence, occlusion flags, calibration quality, and trial validity were recorded. A stratified quality-control subset of 120 unique trials underwent manual review of selected event frames and derived biomechanical values. The validation log included

the ball-contact frame, the support-foot-contact frame, kicking-foot velocity, knee flexion at ball contact, pelvis rotation, and support-foot-to-ball distance. Agreement was evaluated against prespecified tolerances and summarized using absolute error and within-tolerance proportions. This exercise was designed as operational quality control rather than a full criterion-validity or test–retest reliability study; within-tolerance proportions are therefore interpreted as descriptive QC and not as substitutes for ICC, SEM, MDC, or Bland–Altman agreement estimates. The QC subset was designed for operational error checking rather than formal reliability estimation; knee-extension angular velocity was not included as a separate manually validated item. Accordingly, ICC, SEM, relative-error, and Bland–Altman estimates are not reported retrospectively for these velocity variables.

2.9. Statistical Analysis

Descriptive results were summarized at participant and trial levels. Because each player contributed repeated kicks, the primary inferential analysis used generalized estimating equations (GEE) with a Gaussian family, an identity link, an exchangeable within-player working correlation, and robust standard errors. Kicking-leg status (dominant/non-dominant), task condition (accuracy-priority/velocity-priority), and their interaction were entered as fixed effects. Effect estimates are reported with 95% confidence intervals. A prespecified sensitivity analysis repeated the same GEE models after excluding players recorded as having current lower-limb pain or a lower-limb injury within the preceding three months. This sensitivity analysis was not used to estimate an injury effect or as a machine-learning predictor; it assessed whether the primary condition contrasts were robust to removal of flagged participants. The continuous primary outcomes were radar-measured ball velocity and radial target error. For the sensitivity analysis, the current-pain flag denoted current lower-limb pain and the recent-injury flag denoted a lower-limb injury within the preceding three months among participants who otherwise met the eligibility and clearance criteria. Ascertainment source, laterality relative to the dominant leg, diagnosis, and performance impact were not available as standardized analysis fields; the flags were therefore used only for sensitivity exclusion and were not interpreted as injury effects.

2.10. Machine-Learning Analysis

The two prediction tasks were ball velocity and radial target error, selected a priori to represent mechanical output and spatial precision, respectively. Elastic Net regression served as the transparent regularized linear comparator, with radial-basis support vector regression (RBF-SVR), random forest, and gradient-boosted trees as nonlinear comparators. Model selection and evaluation used nested five-fold cross-validation grouped by participant. In each outer split, all trials from held-out participants were excluded from preprocessing, model fitting, and hyperparameter tuning; tuning was performed only within the remaining training participants using three-fold participant-grouped cross-validation. Scaling and all model-specific preprocessing were estimated within training folds. The same outer folds were used for every candidate model. No imputation was required because the 12 prespecified predictors and both primary outcomes were complete for all 1134 valid trials. Performance was summarized using MAE, RMSE, and R^2 from pooled out-of-fold predictions. Ninety-five percent confidence intervals were estimated by nonparametric resampling of participants rather than individual kicks. Model selection followed a prespecified parsimony rule: when predictive performance was materially equivalent, the simpler and more interpretable model was preferred. The locked comparison did not include a single-feature model or an unregularized ordinary least-squares model; therefore, H3 as originally worded could be evaluated only partially. The model comparison addressed

the narrower question of whether nonlinear multivariable models improved on the Elastic Net comparator.

2.11. Privacy and Video Governance

Participant names were not included in the analytic dataset, and raw video was treated as potentially identifiable data. Raw videos were retained separately from the coded analytic dataset, which contained derived kinematic variables and performance outcomes.

2.12. Data Completeness, Final Lock, and Reporting Rules

The outcome and timing audits were completed before the definitive predictive analysis. Binary target hit was recomputed from the physical 1.0 × 1.0 m square boundaries, while continuous radial error remained the primary accuracy outcome. The encoded frame rate of every source video was verified, and mixed-frame-rate views were synchronized using file-specific timing followed by resampling to a common time base. The planned 20 kicks per participant described in Table 1 represented the protocol target; after quality exclusions, 1134 valid trials remained, with 17–20 valid trials per participant. The 12 pre-specified biomechanical predictors listed in Table 2 were complete for all 1134 retained valid trials. No outcome or feature definition was changed after the final model-performance results were examined.

Table 1. Pre-specified field protocol.

Element	Pre-Specified Decision
Participants	60 U16–U19 football players
Warm-up	10–15 min: light running, dynamic mobility, progressive accelerations, and 4–6 submaximal practice kicks
Distance	11 m from stationary ball to goal
Trials	20 planned kicks/player: 5 dominant-velocity, 5 dominant-accuracy, 5 non-dominant-velocity, 5 non-dominant-accuracy; final retained valid trials ranged 17–20/player after quality exclusions
Order	Four five-kick blocks; block order randomized using a fixed pre-generated schedule
Recovery	45 s between kicks; 2 min between blocks
Accuracy target	1.0 × 1.0 m contralateral lower-corner target; center 1.0 m inside the post and 0.75 m above ground
Primary outcomes	Ball velocity (m/s) and radial target error (cm)
Video	Four synchronized cameras, preferably 1080p at ≥120 frames/s
AI pipeline	Multiview pose estimation and 3D triangulation; model/software version locked after a separate pilot
Validation	Nested 5-fold participant-wise cross-validation; all trials from one player remain in the same fold

Table 2. Primary biomechanical predictor set.

Feature	Primary Event/Window	Unit
Approach angle	Final approach to support-foot contact	degrees
Approach velocity	Final approach	m/s
Hip flexion/extension	Ball contact	degrees
Knee flexion	Ball contact	degrees
Ankle angle	Ball contact	degrees
Peak knee-extension angular velocity	Pre-contact swing	degrees/s

Table 2. *Cont.*

Feature	Primary Event/Window	Unit
Kicking-foot linear velocity	Immediately before ball contact	m/s
Pelvis rotation	Ball contact	degrees
Trunk sagittal inclination	Ball contact	degrees
Trunk lateral inclination	Ball contact	degrees
Support-foot-to-ball distance	Ball contact	cm
Support-foot orientation	Ball contact	degrees

2.13. Model Specification and Selection

The search space was fixed before the definitive model comparison with the candidate models, their roles, key tuning parameters, and primary reporting metrics summarized in Table 3. Elastic Net tuned the regularization strength and L1 ratio; RBF-SVR tuned C, gamma, and epsilon; random forest tuned maximum depth, minimum samples per leaf, and the fraction of predictors considered at each split; and gradient boosting tuned learning rate, tree depth, and number of estimators. The optimization criterion in the inner loop was MAE. Gradient boosting achieved the lowest numerical MAE for ball velocity, but its advantage over Elastic Net was only 0.014 m/s and the participant-bootstrap 95% CI for the paired MAE difference included zero. Similar out-of-fold performance across Elastic Net and the tree ensembles does not demonstrate that all biomechanical relationships are intrinsically linear; rather, within this sample and feature set, nonlinear interaction structure did not provide measurable incremental predictive performance. Elastic Net was therefore retained as the primary model under the prespecified parsimony rule.

Table 3. Model-comparison framework used in the nested cross-validation analysis.

Model	Role	Key Tuning Parameters	Primary Reporting
Elastic-net regression	Transparent baseline	alpha; L1 ratio	MAE, RMSE, R^2
RBF support vector regression	Nonlinear comparator	C; gamma; epsilon	MAE, RMSE, R^2
Random forest	Tree ensemble	trees; max depth; min samples leaf; max features	MAE, RMSE, R^2
Gradient-boosted trees	Boosted nonlinear model	learning rate; depth; estimators; subsampling	MAE, RMSE, R^2

Note: MAE = mean absolute error; RMSE = root mean square error; R^2 = coefficient of determination; RBF = radial basis function; L1 = least absolute shrinkage and selection operator regularization component.

For both outcomes, candidate models were compared using pooled participant-wise out-of-fold MAE, supported by RMSE and R^2. Confidence intervals were generated by participant-level bootstrap resampling. Pairwise model differences were evaluated on the same held-out predictions, preserving the participant grouping used during cross-validation.

2.14. Explainability, Feature Stability, and Reproducibility

Because the transparent Elastic Net model performed equivalently to the best nonlinear models for ball velocity, interpretation prioritized standardized model coefficients and participant-wise outer-fold permutation importance. Feature stability was assessed across the five outer folds by recording whether each variable ranked among the five strongest predictors. Permutation importance was calculated only on held-out outer-fold data. Explanations were interpreted as descriptions of predictive model behavior rather than causal effects.

Statistical analyses (descriptive summaries, GEE, and sensitivity analyses) were conducted in Python 3.13.5 using NumPy 2.3.5, pandas 2.2.3, SciPy 1.17.0, and statsmodels 0.14.6. Machine-learning analyses used scikit-learn 1.8.0, and visualizations used Matplotlib 3.10.8; random seed 2026 was used where applicable. Markerless reconstruction used Pose2Sim v0.6.1 with the OpenPose BODY_25 model. The participant-wise outer-fold assignments, feature definitions, quality-control tolerances, and model hyperparameters are provided in the Supplementary Materials. Raw identifiable video is not intended for unrestricted release; derived de-identified data and analysis code may be made available subject to institutional ethics and data-protection requirements.

3. Results

3.1. Participants and Trial Retention

The cohort comprised 60 players (48 male, 12 female). Mean age at the kicking session was 16.35 ± 1.19 years (range 14.31–18.28) as presented in Table 4. The dataset contained 1252 attempted kicks, of which 1134 (90.6%) met the validity criteria; the 118 invalid attempts corresponded to an average of 1.97 invalid attempts per participant. Participants contributed 17–20 valid trials each; 21 players completed all 20 planned valid trials. All 1134 valid trials had a corresponding biomechanics record.

Table 4. Participant characteristics.

Characteristic	Value
Participants	60
Sex	48 male; 12 female
Age at testing, years	16.35 ± 1.19 (14.31–18.28)
Height, cm	173.6 ± 7.9
Body mass, kg	67.7 ± 7.6
Training age, years	8.1 ± 2.4
Weekly training, h	9.8 ± 2.4
Dominant kicking leg	52 right; 8 left

3.2. Kicking Performance by Task and Leg

Velocity-priority trials produced higher ball speed and larger radial error than accuracy-priority trials. Dominant-leg trials were faster and more accurate than non-dominant-leg trials. Binary target-hit percentages below use the corrected 1.0 × 1.0 m square-target definition (see Table 5). Participant-level speed–accuracy profiles across task conditions and kicking legs are presented in Figure 2.

Table 5. Trial-level kicking performance by task condition and leg.

Condition	Leg	Valid n	Ball Speed (m/s)	Radial Error (cm)	Square-Target Hit
Accuracy	Dominant	292	26.60 ± 2.61	35.60 ± 17.14	86.3%
Accuracy	Non-dominant	287	23.47 ± 2.62	39.93 ± 17.02	81.2%
Velocity	Dominant	282	28.95 ± 2.65	57.10 ± 20.83	48.6%
Velocity	Non-dominant	273	25.92 ± 2.86	66.30 ± 19.30	32.2%

Values are mean ± SD for valid trial-level observations. These descriptive observations are repeated within participants and should not be interpreted as independent-sample inferential estimates.

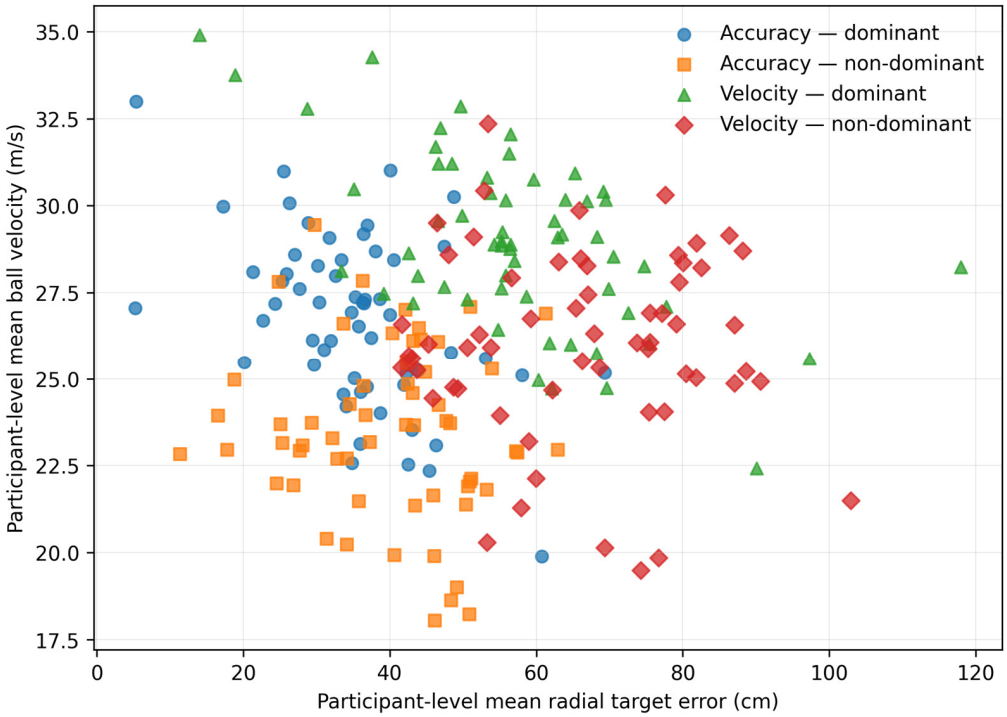


Figure 2. Participant-level speed–accuracy profiles by task condition and kicking leg. Each point represents a participant-specific mean across valid trials. The figure is based on radar ball velocity and continuous radial target error. It is descriptive; no condition-specific regression line was prespecified.

3.3. Participant-Clustered Associations

Participant-clustered GEE results showed that, relative to dominant-leg accuracy-priority trials, the non-dominant leg was associated with lower ball velocity, while the velocity-priority instruction increased ball velocity. For radial error, both non-dominant-leg use and velocity-priority instruction increased error, with an additional positive interaction. The detailed participant-clustered GEE estimates for ball velocity and radial target error are presented in Table 6.

Table 6. Participant-clustered GEE estimates for ball velocity and radial target error.

Outcome	Effect	Estimate	95% CI	*p*
Ball velocity (m/s)	Non-dominant leg	−3.16	−3.71 to −2.60	<0.001
Ball velocity (m/s)	Velocity-priority task	+2.30	+2.12 to +2.48	<0.001
Ball velocity (m/s)	Leg × task	+0.19	−0.08 to +0.46	0.162
Radial error (cm)	Non-dominant leg	+4.52	+0.96 to +8.09	0.013
Radial error (cm)	Velocity-priority task	+21.76	+19.01 to +24.51	<0.001
Radial error (cm)	Leg × task	+4.65	+0.89 to +8.40	0.015

Note: GEE = generalized estimating equations; CI = confidence interval; *p* = probability value; m/s = meters per second; cm = centimeters. Negative estimates indicate lower values relative to the reference condition, whereas positive estimates indicate higher values.

Sensitivity Analysis

The sensitivity analysis excluded 14 players with a current-pain or recent-injury flag, leaving 46 participants and 874 valid trials. The principal performance pattern was preserved: the non-dominant leg remained approximately 3.05 m/s slower and the velocity-priority condition approximately 2.34 m/s faster for ball velocity. For radial error, the velocity-priority condition remained strongly associated with greater error (+21.30 cm),

while the leg main effect was attenuated ($p = 0.052$) and the leg × task interaction remained significant (+5.60 cm, $p = 0.013$).

3.4. Markerless Quality Control

Manual quality control covered 120 unique trials and 167 individual checks. The item-level quality-control results, including mean absolute errors and within-tolerance proportions, are presented in Table 7. Overall, 157 of 167 checks (94.0%) were within the predefined tolerance. Ball-contact-frame localization was the weakest item: 78.6% of checks were within tolerance, with a mean absolute error of 1.18 frames. By comparison, support-foot-contact localization, kicking-foot velocity, and pelvis rotation each met tolerance in all reviewed checks. These results document operational QC but do not constitute a full criterion-validity or test–retest reliability analysis. Knee-extension angular velocity did not have a separate manual-validation check in this QC subset.

Table 7. Markerless quality-control results.

Validation Item	n	Mean Absolute Error	Within Tolerance
Ball-contact frame	28	1.18 frames	78.6%
Support-foot-contact frame	30	1.00 frames	100%
Kicking-foot velocity	27	0.34 m/s	100%
Knee flexion at contact	21	2.24°	85.7%
Pelvis rotation	30	2.10°	100%
Support-foot-to-ball distance	31	1.22 cm	96.8%

3.5. Data-Audit Resolution

The corrected 1.0 × 1.0 m square-target definition altered the binary classification of 93 of 1134 valid kicks relative to the earlier radial-threshold label; continuous radial error was unchanged. Source-file metadata verified the encoded FPS for all 5008 video files. Many trials contained both 120- and 240-frame/s views, but each source file's own timing information was used, and views were resampled/synchronized to a common time base before multiview reconstruction. The file-level timing audit therefore resolved the earlier discrepancy, and the 12 prespecified time-derived and event-specific biomechanical predictors were retained for the locked analysis.

3.6. Final Participant-Wise Nested Cross-Validation

Definitive five-fold participant-wise nested cross-validation showed relatively strong internal predictive performance for ball velocity but minimal predictive performance for radial target error. For ball velocity, Gradient Boosting achieved the numerically lowest MAE (1.17 m/s), but its improvement over Elastic Net was only 0.014 m/s and the participant-bootstrap 95% CI for the paired MAE difference included zero (−0.055 to +0.026 m/s). Elastic Net was therefore retained as the primary model under the prespecified parsimony rule. Its out-of-fold performance in participants excluded from model fitting and tuning was an MAE of 1.19 m/s (95% CI 1.12–1.25), an RMSE of 1.49 m/s (95% CI 1.41–1.56), and an R^2 of 0.80 (95% CI 0.76–0.83). The relationship between observed and out-of-fold predicted ball velocity for the selected Elastic Net model is illustrated in Figure 3. For radial error, all models remained close to the mean-prediction baseline; Elastic Net yielded an MAE of 17.78 cm (95% CI 17.00–18.60), an RMSE of 22.22 cm (95% CI 21.22–23.30), and an R^2 of 0.01 (95% CI −0.02 to 0.04).

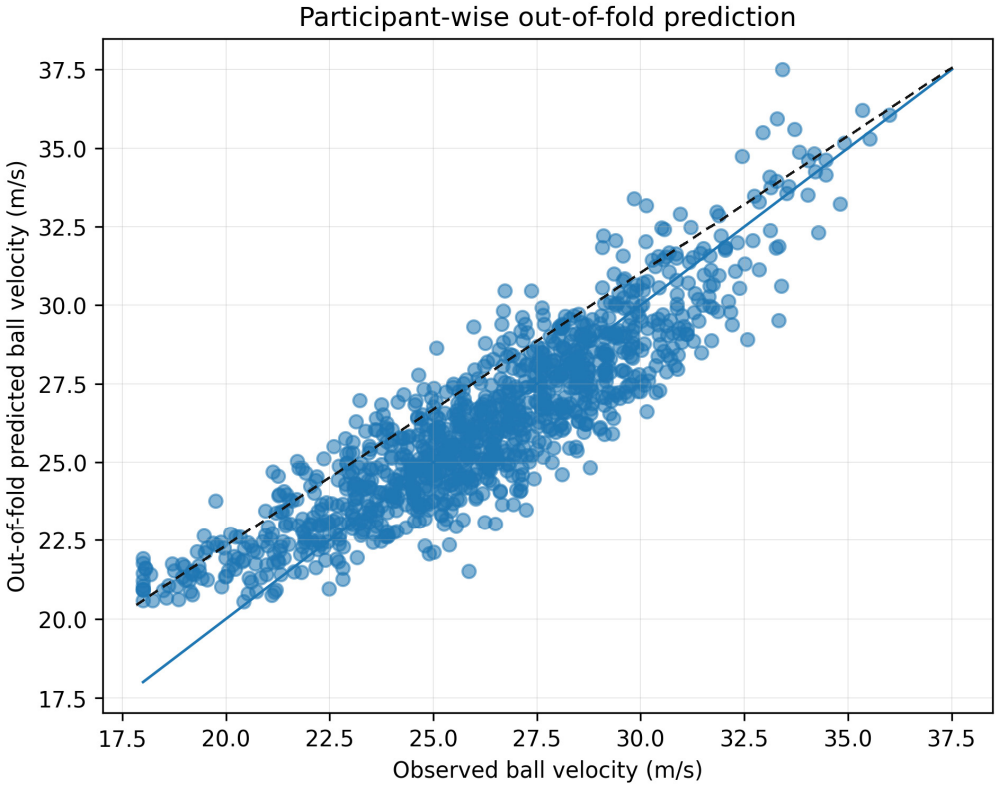


Figure 3. Observed versus out-of-fold predicted ball velocity for the selected Elastic Net model. Predictions were generated only for participants held out from model fitting and inner-loop tuning. The dark dashed line indicates the line of identity (predicted = observed).

3.7. Model Comparison and Model Selection

Across the four prespecified models, ball-velocity prediction was consistently strong for Elastic Net, Random Forest, and Gradient Boosting, whereas RBF-SVR performed slightly worse. Gradient Boosting achieved the lowest numerical MAE (1.17 m/s), but the paired participant-bootstrap comparison with Elastic Net showed no clear advantage. Because the prespecified selection rule favored the simpler model when performance was comparable, Elastic Net was selected as the primary model. This equivalence should not be interpreted as proof of an intrinsically linear biomechanical system; it indicates only that the tested nonlinear models did not extract additional out-of-fold signal from these 12 predictors in this sample. For radial target error, no model achieved meaningful out-of-fold explanatory power. The participant-wise nested cross-validation results for all four predictive models, including MAE, RMSE, R^2, and 95% confidence intervals, are summarized in Table 8.

Table 8. Participant-wise nested cross-validation results.

Outcome	Model	MAE	RMSE	R^2	Participant-Level 95% CI
Ball velocity	Elastic net	1.19	1.49	0.80	MAE 1.12–1.25; R^2 0.76–0.83
Ball velocity	RBF-SVR	1.26	1.59	0.77	MAE 1.20–1.33; R^2 0.73–0.80
Ball velocity	Random forest	1.18	1.48	0.80	MAE 1.13–1.25; R^2 0.75–0.84
Ball velocity	Gradient boosting	1.17	1.47	0.80	MAE 1.11–1.23; R^2 0.76–0.84
Radial error	Elastic net	17.78	22.22	0.01	MAE 17.00–18.60; R^2 −0.02–0.04

Table 8. *Cont.*

Outcome	Model	MAE	RMSE	R^2	Participant-Level 95% CI
Radial error	RBF-SVR	17.81	22.35	0.00	MAE 16.96–18.70; R^2 −0.03–0.02
Radial error	Random forest	17.83	22.34	0.00	MAE 17.03–18.67; R^2 −0.03–0.03
Radial error	Gradient boosting	17.88	22.42	−0.00	MAE 17.05–18.75; R^2 −0.03–0.02

Note: MAE = mean absolute error; RMSE = root mean square error; R^2 = coefficient of determination; CI = confidence interval; RBF-SVR = radial basis function support vector regression. All performance values are based on participant-wise out-of-fold predictions from nested cross-validation.

3.8. Explainability and Feature Stability

The Elastic Net model showed highly stable importance for three mechanical variables: kicking-foot velocity, knee-extension angular velocity, and approach velocity ranked among the top five predictors in all five outer folds.

Kicking-foot velocity and knee-extension angular velocity had the largest standardized coefficients (mean absolute coefficients of 1.34 and 1.35, respectively) and the largest held-out permutation importance values. These values are model-scale coefficients after standardization and regularization, not raw mechanical effect sizes in m/s. Their magnitudes, compared with 0.48 for approach velocity, indicate greater model reliance conditional on the other included predictors rather than a causal ranking of biomechanical determinants. The mean absolute standardized coefficients of the fold-stable predictors in the selected Elastic Net ball-velocity model are illustrated in Figure 4.

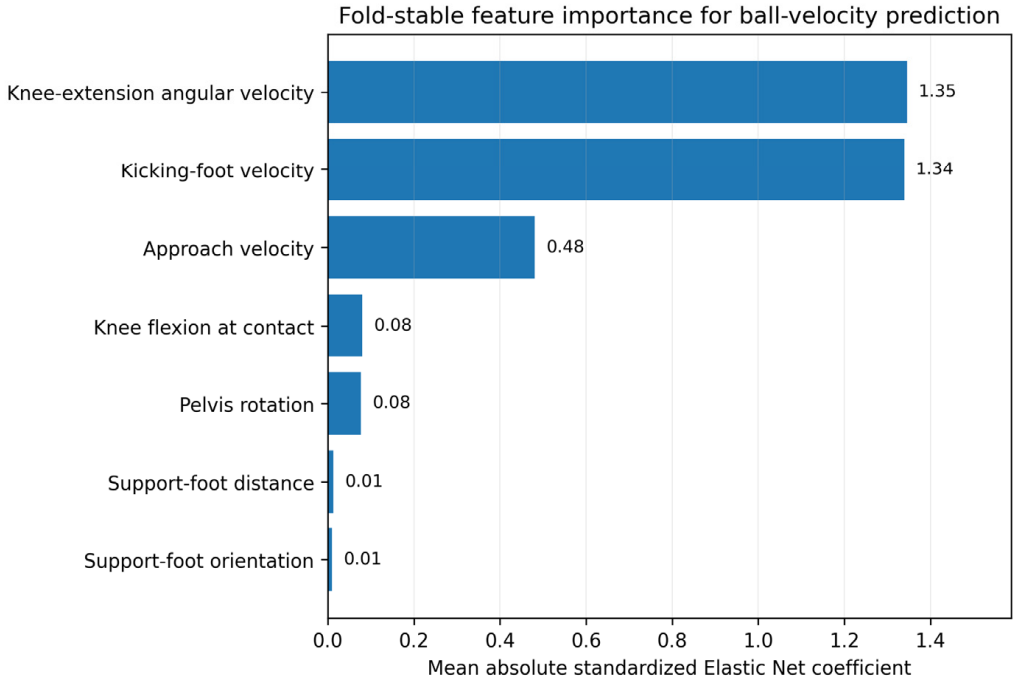


Figure 4. Fold-stable feature importance for the selected Elastic Net ball-velocity model. Bars show mean absolute standardized coefficients across the five outer participant-wise folds.

Knee flexion at ball contact and pelvis rotation were smaller but consistently retained among the top five predictors. he standardized coefficients, permutation importance values, and ranking stability of the five leading predictors of ball velocity are summarized in Table 9. The remaining variables contributed little additional predictive information after regularization.

Because the radial-error models had essentially no out-of-fold explanatory power, feature-importance plots for shooting accuracy are not interpreted as reliable technical determinants. More generally, all feature-importance results describe predictive model behavior and should not be read as causal effects.

Table 9. Fold-stable Elastic Net predictors of ball velocity.

Feature	Mean Standardized Coefficient	Mean Permutation Importance	Top-5 Across Folds
Knee-extension angular velocity	+1.35	0.761	100%
Kicking-foot velocity	+1.34	0.766	100%
Approach velocity	+0.48	0.136	100%
Knee flexion at contact	−0.08	0.006	100%
Pelvis rotation	+0.08	0.003	100%

4. Discussion

4.1. Principal Performance Findings

The dominant performance pattern was a task-dependent speed–accuracy trade-off. Velocity-priority instruction increased ball velocity but substantially increased radial target error. This direction agrees with experimental football work showing that instructions emphasizing speed can increase ball velocity while reducing accuracy, particularly for demanding target locations [1,2].

The non-dominant leg was slower and less accurate. This finding is best interpreted as a within-player laterality effect rather than evidence that leg dominance itself causes the difference. The comparison is relevant to applied coaching because players are frequently required to finish or pass under constraints that prevent use of the preferred limb.

The leg × task interaction was small and non-significant for ball velocity but positive and significant for radial error. This pattern suggests that the cost of velocity emphasis was expressed more strongly in precision control than in the ability to increase speed.

4.2. Biomechanical Interpretation

Kicking-foot velocity and knee-extension angular velocity emerged as the strongest mechanical predictors of ball velocity, followed by approach velocity. This pattern is compatible with prior youth work showing age- and status-related differences in lower-extremity kicking mechanics [3] and with professional-player analyses linking hip and knee movement patterns to kicking velocity [4].

No single joint variable should be interpreted as the cause of performance. The kick is a coordinated whole-body action, and the coefficient and permutation-importance analyses describe model reliance rather than causal effects.

Support-foot geometry, trunk lean, and pelvis rotation did not provide sufficient information for meaningful prediction of radial target error. Variables such as gaze, perception, approach consistency, foot–ball contact location, and ball-impact mechanics could plausibly contribute to precision control, but they were not measured here. These possibilities are therefore hypothesis-generating explanations, not mechanisms established by the present data.

4.3. Markerless Measurement Quality

Quality-control results were encouraging overall, with 94.0% of checks within prespecified tolerance, but ball-contact-frame localization remained the principal temporal source of error. Only 78.6% of reviewed ball-contact frames were within tolerance, with a mean

absolute error of 1.18 frames; a one- or two-frame shift can affect event-specific foot or joint velocity during high-speed kicking. This limitation could not be eliminated retrospectively and should be considered when interpreting event-level velocity features. The reported out-of-fold R^2 of 0.80 should therefore be interpreted in light of temporal uncertainty in two leading time-derived predictors—kicking-foot velocity and knee-extension angular velocity—and the absence of an independent manual-validation check for knee-extension angular velocity.

The field workflow therefore demonstrates feasibility rather than equivalence to laboratory optical motion capture. Positional and angular agreement should be distinguished from time-derived velocity reliability. The file-level frame-rate audit and common-time-base resampling reduce timing inconsistency, but 120-frame/s views contain less temporal information than 240-frame/s views, and resampling cannot recover information that was not acquired.

This qualification is consistent with the broader markerless literature, which emphasizes that performance depends on view geometry, occlusion, movement speed, calibration, and the downstream variable being estimated [6,7,19]. OpenPose BODY_25 may be particularly challenged by rapid rotations, self-occlusion, motion blur, and high distal-segment velocities; multiview triangulation can reduce but cannot fully remove upstream 2D keypoint error.

4.4. Machine Learning and Explainability

The definitive participant-wise nested analysis showed relatively strong internally validated performance for estimating ball velocity, whereas radial target error remained poorly estimated. This asymmetry is more informative than a single aggregate AI-performance score and should not be generalized beyond the present internally validated cohort.

One hypothesis-generating interpretation is that mechanical output is more directly encoded in observable segment kinematics, whereas precision may involve additional motor-control and perceptual variables. Because those variables were not measured, the near-baseline radial-error performance should be treated as an empirical finding, not as evidence for a specific unmeasured mechanism.

Participant-wise nested cross-validation is a methodological strength because all kicks from a given player remained together. This reduces identity leakage and provides a more credible estimate of performance for an unseen player from the same population [16,17].

Explainability was used to inspect model behavior, not to convert prediction into causal coaching prescriptions. With respect to the prespecified hypotheses, H2 was not supported because the selected support-foot, trunk, and pelvis variables did not explain radial target error; H3 was only partially evaluated: added nonlinear complexity did not materially improve out-of-fold performance over Elastic Net, but the absence of single-feature and unregularized ordinary least-squares comparators precludes a full test of the original hypothesis. The exploratory compact-feature expectation was supported for ball velocity but not for radial error. Any technical recommendation based on these features should remain hypothesis-generating until tested prospectively.

4.5. Practical Relevance for Football

The workflow is potentially useful as a descriptive assessment workflow in academy environments because it combines cameras, radar, and automated pose analysis to produce player-level information without laboratory markers.

A practical report could separate a player's power-related profile from precision-related performance. For example, a player may produce high foot and ball velocity yet show large radial error under velocity-priority instruction, whereas another may maintain

accuracy but produce lower ball speed. Such profiles may help structure assessment questions without allowing the algorithm to make selection decisions.

The youth passing and technology-assisted training literature supports the broader principle that technical and perceptual components can be trained [10,12,13], but the present field experiment cannot establish that changing an AI-identified feature will improve kicking performance.

4.6. Limitations and Next Study

The independent sample comprised 60 players, not 1134 independent kicks. This is a modest independent sample for participant-wise comparison of four predictive models and feature-stability analysis. Repeated trials improve precision for within-player estimates but do not establish generalizability beyond the study population.

The sex distribution was unbalanced (48 male, 12 female), and sex was not included as a predictive covariate because the female subgroup was too small for stable participant-wise stratified modeling. Sex-associated differences in outcomes or biomechanics therefore cannot be excluded. Laterality was also imbalanced (52 right- and 8 left-dominant players), so dominant/non-dominant findings should not be generalized to specific right- versus left-foot mechanics.

The models were evaluated only by internal participant-wise nested cross-validation; no independent external cohort was available. The standardized 11 m stationary-ball task, the Romanian youth-program recruitment context, unequal numbers of valid trials, and current-pain or recent-injury flags in a subset of medically cleared participants further limit transportability. The sensitivity analysis excluding flagged players preserved the principal findings, but external validation is required before broader use.

Measurement limitations include heterogeneous 120/240-frame/s source recordings, the inability of resampling to restore temporal detail absent from lower-frame-rate views, imperfect ball-contact-frame localization, and the sensitivity of OpenPose-based keypoint estimation to rapid rotation and occlusion. Residual inter-camera synchronization error was not independently quantified. In addition, perceptual-cognitive variables such as gaze, scanning, and reaction measures and direct foot–ball contact variables were not assessed; proposed roles for these factors in shooting accuracy therefore remain hypotheses. Future work should prospectively test the locked model in an independent academy or club and add such variables only under a prespecified validation plan. Future validation should include paired manual/reference checks of ball-contact timing, kicking-foot velocity, and knee-extension angular velocity with prespecified absolute/relative error and agreement statistics where the design supports them.

4.7. Positioning Against Recent Football-Specific Markerless Evidence

Recent football-specific validation work has shown that markerless pose-estimation systems can be useful for screening movement trends during long-pass tasks, while high-precision agreement with inertial reference systems can vary across joints and algorithms [14]. This supports interpreting the present system as a field-performance assessment workflow rather than a laboratory-equivalent motion-capture replacement.

The wider markerless sports literature emphasizes ecological validity and accessibility as major strengths, while camera geometry, occlusion, model choice, calibration, and movement speed remain important determinants of error [19]. Quality control is therefore central to interpreting field-based markerless outputs rather than a minor technical detail. Recent work also shows how deep learning and machine learning can extend sport analysis by automating pose or motion tracking, recognizing football actions from wearable data, and extracting complex spatiotemporal features for performance assessment [6,8,9,20].

These advances broaden the analytic toolkit but do not remove the need for task-specific validation and participant-aware evaluation.

The principal contribution is not simply the use of markerless pose estimation, but the combination of field validation, continuous football performance outcomes, participant-aware machine learning, and a direct test of whether the same biomechanical features predict mechanical output and precision equally well.

5. Conclusions

Field-based markerless biomechanics showed relatively strong internally validated performance for estimating ball velocity in held-out youth players, while the same feature set did not meaningfully estimate shooting precision. An interpretable Elastic Net model performed essentially as well as more complex nonlinear alternatives, with kicking-foot velocity, knee-extension angular velocity, and approach velocity providing the most stable predictive signal. The model comparison supports the narrower conclusion that added nonlinear complexity did not materially improve participant-wise out-of-fold performance over Elastic Net; because single-feature and unregularized linear comparators were not included, the original H3 was only partially evaluated. These findings are limited to internal validation and should not be interpreted as evidence of external generalizability or causal technical determinants. Interpretation is also constrained by heterogeneous 120/240-frame/s recordings, imperfect ball-contact-frame localization, and known challenges of OpenPose-based reconstruction during rapid rotation and occlusion. External validation in an independent cohort and prospective evaluation of additional perceptual and ball-contact variables are required before broader coaching or decision-support use.

Supplementary Materials: The following supporting information can be downloaded at https://www.mdpi.com/article/10.3390/app16209986/s1, Table S1: Variables and operational definitions; Table S2: Outer participant-fold allocation; Table S3: Outer-fold model tuning results and selected hyperparameters; Table S4: Markerless quality-control results; Table S5: Camera calibration and acquisition configuration.

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Informed Consent Statement: Written informed consent was obtained from adult participants. For the 54 participants who were minors at testing, written parent/legal-guardian consent and participant assent were documented. A de-identified archive audit confirmed that the study-specific participant information included video recording and AI-based analysis and that consent/assent was dated on or before testing.

Data Availability Statement: De-identified derived data and analysis code are available from the corresponding author on reasonable request, subject to institutional ethics and data-protection requirements.

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References

1. Sørensen, A.; Christensen, O.H.; van den Tillaar, R. Effect of Instruction and Target Position on Penalty Kicking Performance in Soccer. *Appl. Sci.* **2024**, *14*, 10668. [CrossRef]
2. van den Tillaar, R.; Fuglstad, P. Effect of Instructions Prioritizing Speed or Accuracy on Kinematics and Kicking Performance in Football Players. *J. Mot. Behav.* **2017**, *49*, 414–421. [CrossRef] [PubMed]
3. Palucci Vieira, L.H.; Barbieri, F.A.; Kellis, E.; Oliveira, L.; Aquino, R.; Cunha, S.; Bedo, B.; Manechini, J.; Santiago, P. Organisation of Instep Kicking in Young U11 to U20 Soccer Players. *Sci. Med. Footb.* **2021**, *5*, 111–120. [CrossRef] [PubMed]
4. Navandar, A.; Kipp, K.; Navarro, E. Hip and Knee Joint Angle Patterns and Kicking Velocity in Female and Male Professional Soccer Players: A Principal Component Analysis of Waveforms Approach. *J. Sports Sci.* **2022**, *40*, 1919–1930. [CrossRef] [PubMed]
5. Palucci Vieira, L.H.; Santiago, P.R.P.; Pinto, A.; Aquino, R.; Torres, R.d.S.; Barbieri, F.A. Automatic Markerless Motion Detector Method against Traditional Digitisation for 3-Dimensional Movement Kinematic Analysis of Ball Kicking in Soccer Field Context. *Int. J. Environ. Res. Public Health* **2022**, *19*, 1179. [CrossRef] [PubMed]
6. Aulton, C.; Wakili, L.; Strafford, B.W.; Davids, K.; Chiu, C.-Y. The Application of Deep Learning Human Pose Estimation in Sport: A Systematic Review. *Sports Med. Open* **2025**, *11*, 155. [CrossRef] [PubMed]
7. Carretier, N.; Beurienne, E.; Beauséjour, M.H.; Gros, L.; Bruna-Rosso, C.; Dorsemaine, M.; Behr, M.; Bailly, N.; Clément, J. Validation of a Markerless Multi-Camera Pipeline for Bouldering Fall Kinematics. *Sensors* **2026**, *26*, 662. [CrossRef] [PubMed]
8. Stoeve, M.; Schuldhaus, D.; Gamp, A.; Zwick, C.; Eskofier, B.M. From the Laboratory to the Field: IMU-Based Shot and Pass Detection in Football Training and Game Scenarios Using Deep Learning. *Sensors* **2021**, *21*, 3071. [CrossRef] [PubMed]
9. Cuperman, R.; Jansen, K.M.B.; Ciszewski, M.G. An End-to-End Deep Learning Pipeline for Football Activity Recognition Based on Wearable Acceleration Sensors. *Sensors* **2022**, *22*, 1347. [CrossRef] [PubMed]
10. Teriș, S.; Lakotos, I.I.; Koronas, V. Training Model Designed to Strengthen and Improve the Pass for Juniors 10–12 Years. *Bull. Transilv. Univ. Braşov. Ser. IX Sci. Hum. Kinet.* **2024**, *17*, 199–206. [CrossRef]
11. Kryeziu, A.R.; Iseni, A.; Teodor, D.F.; Croitoru, H.; Badau, D. Effect of 12 Weeks of the Plyometric Training Program Model on Speed and Explosive Strength Abilities in Adolescents. *Appl. Sci.* **2023**, *13*, 2776. [CrossRef]
12. Badau, D.; Steff, N.; Badau, A.; Stoica, M. Enhancing Lower-Limb Simple and Choice Reaction Time and Spatial Orientation in Junior Basketball Players by Implementing Fitlight Technology in Sports Training. *Percept. Mot. Ski.* **2025**, *132*, 1392–1417. [CrossRef] [PubMed]
13. Badau, D.; Stoica, A.M.; Litoi, M.F.; Badau, A.; Duta, D.; Hantau, C.G.; Sabau, A.M.; Oancea, B.M.; Ciocan, C.V.; Fleancu, J.L.; et al. The Impact of Peripheral Vision on Manual Reaction Time Using Fitlight Technology for Handball, Basketball and Volleyball Players. *Bioengineering* **2023**, *10*, 697. [CrossRef] [PubMed]
14. Wang, D.; Yu, J.; Chen, S.; Yang, J.; Jiang, W.; Gong, Y.; Luo, C. Comparative Analysis of Markerless Motion-Capture Models for Assessing Football Kinematics During 30 m Long-Pass Tasks. *Sensors* **2026**, *26*, 3654. [CrossRef] [PubMed]
15. Lees, A.; Asai, T.; Andersen, T.B.; Nunome, H.; Sterzing, T. The Biomechanics of Kicking in Soccer: A Review. *J. Sports Sci.* **2010**, *28*, 805–817. [CrossRef] [PubMed]
16. Karbalaie, A.; Abtahi, F.; Häger, C.K. Participant-Aware Model Validation for Repeated-Measures Data: Comparative Cross-Validation Study. *JMIR AI* **2026**, *30*, e87728.
17. Collins, G.S.; Moons, K.G.M.; Dhiman, P.; Riley, R.D.; Beam, A.L.; Van Calster, B.; Ghassemi, M.; Liu, X.; Reitsma, J.B.; van Smeden, M.; et al. TRIPOD+AI Statement: Updated Guidance for Reporting Clinical Prediction Models That Use Regression or Machine Learning Methods. *BMJ* **2024**, *385*, e078378. Correction in *BMJ* **2024**, *385*, q902. [CrossRef] [PubMed]
18. Hernández-Belmonte, A.; Sánchez-Pay, A. Concurrent Validity, Inter-Unit Reliability and Biological Variability of a Low-Cost Pocket Radar for Ball Velocity Measurement in Soccer and Tennis. *J. Sports Sci.* **2021**, *39*, 1312–1319. [CrossRef] [PubMed]
19. Bini, R.R.; Colyer, S.; Arruda Moura, F.; Santiago, P.; Vanicek, N. Markerless Motion Analysis in Sports and Exercise. *J. Sports Sci.* **2026**, *44*, 1255–1260. [CrossRef] [PubMed]
20. Jia, Y.; Abdullah, N.A.; Eliza, H.; Lu, Q.; Si, D.; Guo, H.; Wang, W. A Narrative Review of Deep Learning Applications in Sports Performance Analysis: Current Practices, Challenges, and Future Directions. *BMC Sports Sci. Med. Rehabil.* **2025**, *17*, 249. [CrossRef] [PubMed]

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